

How to write an R Markdown file

A short guide, using lecture03_tidyverse.Rmd as the example

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1 What an R Markdown file is

An R Markdown file (`.Rmd`) is a text file that mixes three things:

1. a **header** at the top, between two lines of `---`, that says what the document is called and what it should become (PDF or web page);
2. **text**, written in Markdown, a plain way of marking headings, bold and lists;
3. **code chunks**, blocks of R code that run when the document is *knitted*, with their output placed right after them.

Pressing **Knit** in RStudio turns the file into a finished document: your explanation, your code and its results, together. That is why assignments should use it: the reader sees exactly what you ran and what came out.

This guide is itself an R Markdown file. Open `rmarkdown_guide.Rmd` next to the PDF and compare them; every part described below is visible in the source.

2 Create one

In RStudio: **File** → **New File** → **R Markdown...** Give it a title and your name, choose **PDF** or **HTML**, click **OK**. RStudio opens a file with example content. Delete everything below the header and write your own.

Save it with **Ctrl + S**, in a folder of its own, with any data in a `data/` subfolder next to it. When knitting, R uses the folder of the `.Rmd` file as its working directory, so `read_csv("data/students.csv")` finds the file with no `setwd()`.

3 The three parts, using the lecture file

3.1 The header

The top of `lecture03_tidyverse.Rmd` looks like this:

```
---
title: "Lecture 3: Introduction to the Tidyverse"
author: "Touseef Hameed"
date: "16 September 2026"
output: pdf_document
---
```

Change the title, author and date. `output: pdf_document` makes a PDF; `output: html_document` makes a web page. Keep the indentation exactly as shown; the header is sensitive to spaces.

3.2 Text

Ordinary sentences are just typed. A few marks give structure:

```
# A main heading
## A smaller heading
```

Plain paragraph. **Bold**, *italic*, and `code in the text`.

- a bullet point
- another one

1. a numbered step
2. the next step

A link: <https://r4ds.hadley.nz>

In an assignment, use one `#` heading per problem, then the code, then your explanation.

3.3 Code chunks

A chunk starts with ````{r}` on its own line and ends with `````:

```
```{r}
library(tidyverse) # load the packages
students <- read_csv("data/students.csv") # read the data
head(students) # first six rows
```
```

Insert a new chunk with **Ctrl + Alt + I**. Run one chunk without knitting by clicking the green **play** button at its top right. Its output appears under it in the editor.

When the document is knitted, that chunk produces this (the long message that `library(tidyverse)` prints is hidden here with `message=FALSE`, see Section 4):

```
library(tidyverse)
students <- read_csv("data/students.csv", show_col_types = FALSE)
head(students)
```

```
#> # A tibble: 6 x 5
#>   'Student ID' 'Full Name' favourite.food mealPlan      AGE
#>   <dbl> <chr>          <chr>          <chr>      <chr>
#> 1         1 Sunil Huffmann Strawberry yoghurt Lunch only      4
#> 2         2 Barclay Lynn   French fries    Lunch only      5
#> 3         3 Jayendra Lyne  N/A            Breakfast and lunch 7
#> 4         4 Leon Rossini    Anchovies      Lunch only    <NA>
#> 5         5 Chidiegwu Dunkel Pizza          Breakfast and lunch five
#> 6         6 Güvenç Attila    Ice cream      Lunch only      6
```

4 Chunk options you will need

Options go inside the braces after `r`, separated by commas.

| option | effect | use it for |
|----------------------------|---|---|
| <code>eval=FALSE</code> | show the code, do not run it | <code>install.packages("tidyverse")</code> , which must not run at every knit |
| <code>echo=FALSE</code> | run the code, hide it, show only the output | a plot where the code is not the point |
| <code>include=FALSE</code> | run the code, show nothing | a setup chunk |

| option | effect | use it for |
|---------------------------------|----------------------------|--|
| message=FALSE,
warning=FALSE | hide messages and warnings | the long message that
library(tidyverse) prints |

Example, as written in the lecture file:

```
```{r, eval=FALSE}
install.packages("tidyverse") # run once on each computer you use
```
```

5 Knit

Click **Knit** (the ball-of-wool icon above the editor), or press **Ctrl + Shift + K**. The document is built in a **fresh R session**: nothing from your console is available. Every package must be loaded and every object created *inside* the document, in order.

The result is saved next to the .Rmd file, with the same name and .pdf or .html. That is also how you choose where the PDF goes: save the .Rmd into the folder you want first (**File → Save As...**), for example Documents/oka2015/assignment1/, then knit. If you open the file from Downloads, the PDF lands in Downloads. The Knit button's *Knit Directory* setting changes the working directory, not the output location; leave it at *Document Directory*.

PDF needs a LaTeX installation. If knitting to PDF fails, either run once in the console

```
install.packages("tinytex")
tinytex::install_tinytex()
```

or change the header to output: `html_document`. HTML is accepted for the assignment.

6 The errors everyone meets

| message | cause | fix |
|--|---|---|
| object 'students' not found
could not find function "read_csv" | the chunk that creates it is missing, or comes later in the file
library(tidyverse) is not in the document, or comes after the chunk | create every object before you use it, inside the document
put it in the first chunk |
| cannot open file 'data/students.csv'
the knit stops at install.packages | the file is not in a data folder next to the .Rmd
the chunk is being run | move it there, or use the web address
add eval=FALSE to that chunk |
| Error in parse: unexpected symbol | a typing error in the chunk, often a missing comma or quote | run the chunk alone with the play button to find the line |

The knitting log, in the **R Markdown** tab at the bottom of RStudio, names the chunk where the problem is.

7 For the assignment

- One heading per problem: `# Problem 1`, `# Problem 2`, and so on.
- Under each: the chunk with the code and, for the written problems, your text.
- A comment on every command.
- Knit it, check that the output under each chunk is what you meant, and upload the `.Rmd` together with the knitted `.pdf` or `.html`.

If knitting does not work on your computer, screenshots of your RStudio window showing the code and its output are accepted instead.